

DMS-02 DIE MANAGEMENT OPERATING SYSTEM

WIRE DRAWING TECHNICAL DATA SHEET (TDS) & DRAFTING SCHEDULE

Industrial Pass Progression, Deformation Mechanics & Elongation Profile

DOCUMENT REFERENCE

TDS-WD2026-08P

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WORK ORDER:
WDC-JOB-2026-08P

DRAWING MACHINE:
Multi-Wire Drawing Line 01

MATERIAL PROFILE:
Standard Wire Drawing

QUALITY STATUS:
EXCELLENT (5★ VERIFIED)

01. DRAFTING SCHEDULE & SERIES SUMMARY

8 Drawing Passes Total

INLET SIZE

Ø 3.500 mm

FINISH SIZE

Ø 1.345 mm

OVERALL AREA RED.

85.23 %

TOTAL RATIO

6.77

AVG ELONGATION

27.01 %

AVG AREA RED.

21.27 %

COMPLETE DIE SERIES SEQUENCE:

Ø 3.500

→

Ø 3.100

→

Ø 2.750

→

Ø 2.440

→

Ø 2.165

→

Ø 1.920

→

Ø 1.705

→

Ø 1.515

→

Ø 1.345

02. SCHEMATIC DRAFTING PIPELINE

Sequential cross-section wire reduction and deformation taper through tungsten carbide dies

9 STATIONS • 8 DRAFT PASSES

DRAFTING STAGE 1 (PASSES 1 - 5)

START #1 Ø 3.500
mm
diameter

-21.6%
+27.5%

STATION #2 Ø 3.100
mm
diameter

-21.3%
+27.1%

STATION #3 Ø 2.750
mm
diameter

-21.3%
+27.0%

STATION #4 Ø 2.440
mm
diameter

-21.3%
+27.0%

STATION #5 Ø 2.165
mm
diameter

DRAFTING STAGE 2 (PASSES 6 - 8)

STATION #5 Ø 2.165
mm
diameter

-21.4%
+27.2%

STATION #6 Ø 1.920
mm
diameter

-21.1%
+26.8%

STATION #7 Ø 1.705
mm
diameter

-21.0%
+26.6%

STATION #8 Ø 1.515
mm
diameter

-21.2%
+26.9%

FINISH #9 Ø 1.345
mm
diameter

03. ELONGATION & REDUCTION ENGINEERING DATA TABLE

UNITS: mm, mm², %

Pass-by-pass dimensional reduction and cross-sectional transformation schedule

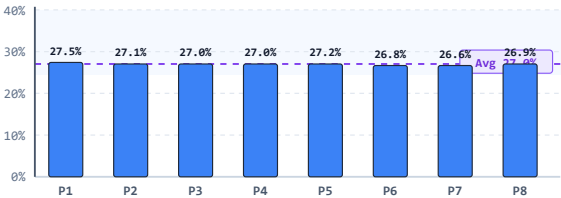
PASS	FROM (Ø mm)	TO (Ø mm)	AREA IN (mm²)	AREA OUT (mm²)	AREA RED. ΔA (%)	ELONGATION E (%)	RATIO R
P1	3.500	3.100	9.621	7.548	21.55 %	27.47 %	1.27
P2	3.100	2.750	7.548	5.940	21.30 %	27.07 %	1.27
P3	2.750	2.440	5.940	4.676	21.28 %	27.03 %	1.27
P4	2.440	2.165	4.676	3.681	21.28 %	27.03 %	1.27
P5	2.165	1.920	3.681	2.895	21.35 %	27.15 %	1.27
P6	1.920	1.705	2.895	2.283	21.14 %	26.81 %	1.27
P7	1.705	1.515	2.283	1.803	21.03 %	26.63 %	1.27
P8	1.515	1.345	1.803	1.421	21.19 %	26.89 %	1.27
AVG	—	—	—	—	21.27 %	27.01 %	1.27
RANGE	—	—	—	—	21.0% - 21.6%	26.6% - 27.5%	—
TOTAL	3.500	1.345	9.621	1.421	85.23 %	577.0 %	6.77

04. GRAPHICAL PASS ANALYSIS

High-contrast vector profiles with mean drafting reference lines and process tolerance limits

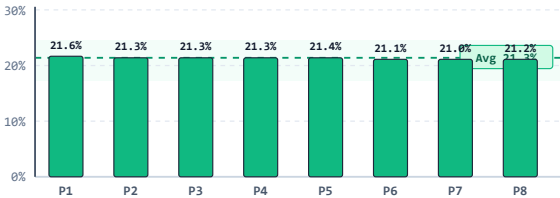
ELONGATION PER PASS (%)

AVG: 27.01% | MAX: 27.47% | MIN: 26.63%



AREA REDUCTION PER PASS (%)

AVG: 21.27% | MAX: 21.55% | MIN: 21.03%



05. OPERATIONAL NOTES & ENGINEERING SIGN-OFF

SHOPFLOOR NOTES & INSTRUCTIONS:

Drafting schedule verified within ±0.5% elongation consistency tolerance. Tungsten carbide die profiles conform to standard drafting geometry. Capstan cooling pressure checked. Wire surface finish inspection required after pass 8.

PREPARED BY:

VERIFIED BY:

QUALITY APPROVAL:

Tool Room Engineer

Production Supervisor

QA / Metallurgist